

WMO-CGS-Workshop, Ethiopia 16 September 2026
WMO Data Interoperability,
Standardization and Exchange

- **TT-W2FH and Other Partners**



HYDROLOGICAL
OBSERVING
SYSTEM



Join WHOS – Participate in the
WHOS Implementation



Interoperability: A key Barriers to Data Exchange

Aspect of interoperability

Technical interoperability

Syntactic interoperability

Semantic interoperability

Organizational interoperability

Legal/regulatory
interoperability



Limited data access, usability, sharing



Data Heterogeneity

Diverse data sources, standards, formats, and scales, vocabularies



Technical Barriers

Complex dimensionality, metadata inconsistencies, limited platforms and interoperability



Quality & Capacity

Insufficient resources, data quality issues, insufficient expertise



Policy & Governance

Data policy restrictions, limited frameworks, lack of awareness



Application & Users

Disconnect with decision makers, challenges in real-world application

WHOS and Interoperability

Multiple partners work to support hydrological data sharing

- Discoverable, accessible, and interoperable.

Adopt standards and interoperability

- “Interoperability enables information systems to do more with the same data—which can change lives.” — Chris Lawrence

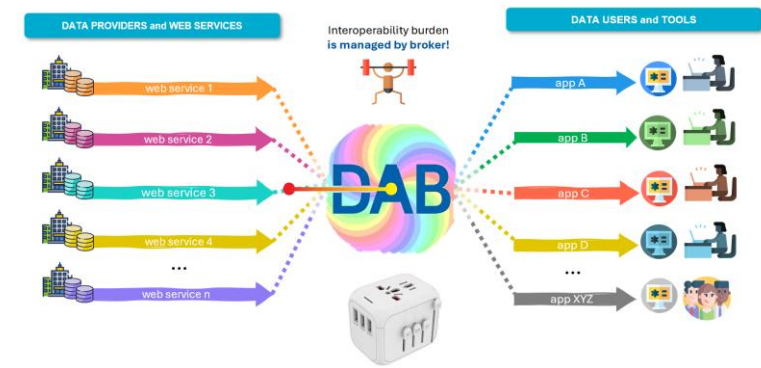
Data sharing: unlocking the power of hydrological data

- Empowering people, enabling applications, informing policy, protecting the environment

WHOS

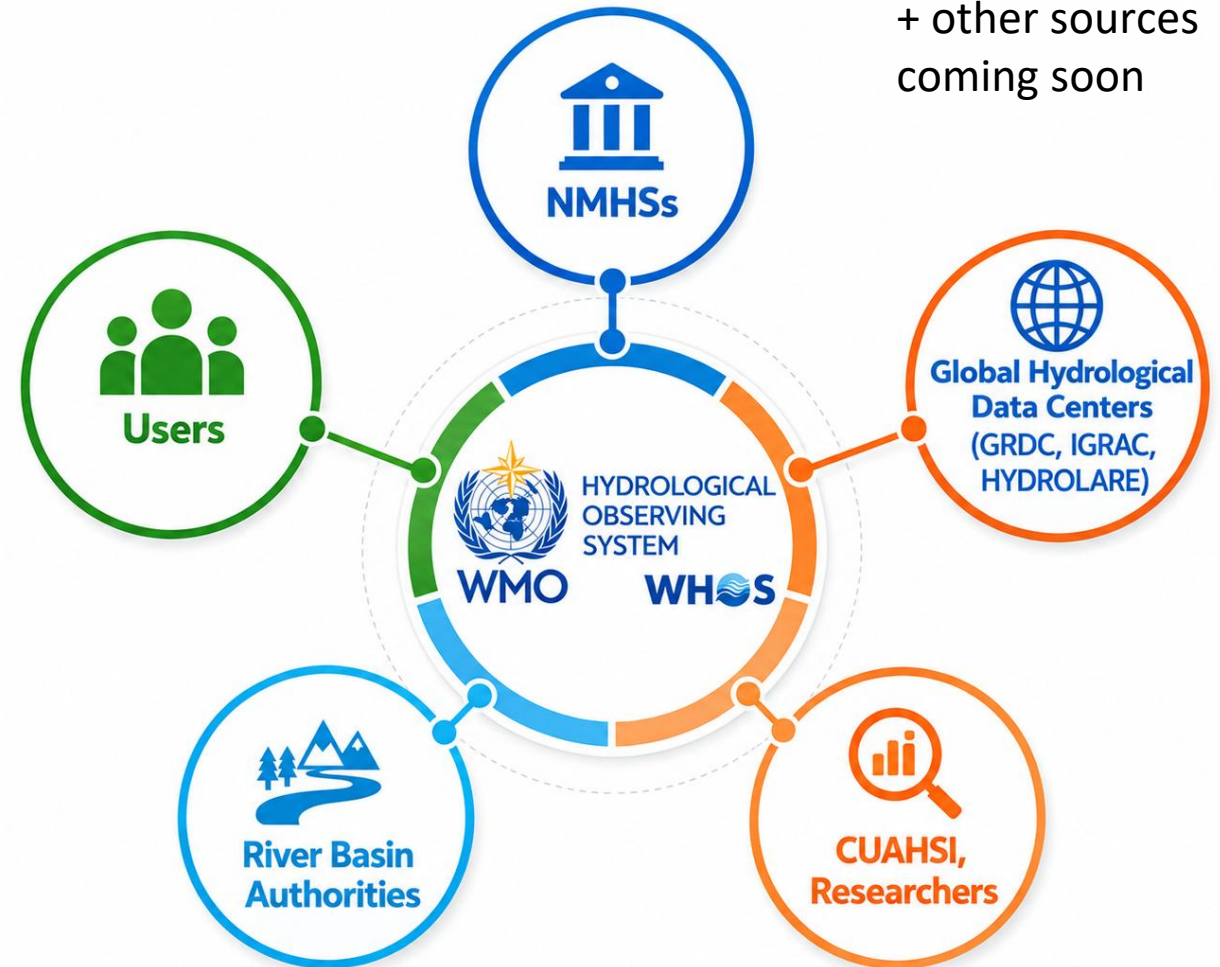
- One platform, multiple providers, Unlimited discovery

WHOS (WIS2 component):
A System of Systems that supports interoperable hydrological data discoverability, harmonizes complex hydrological data, **and provides** a single point for accessing hydrology data and visualizing hydrological data (connecting the dots).



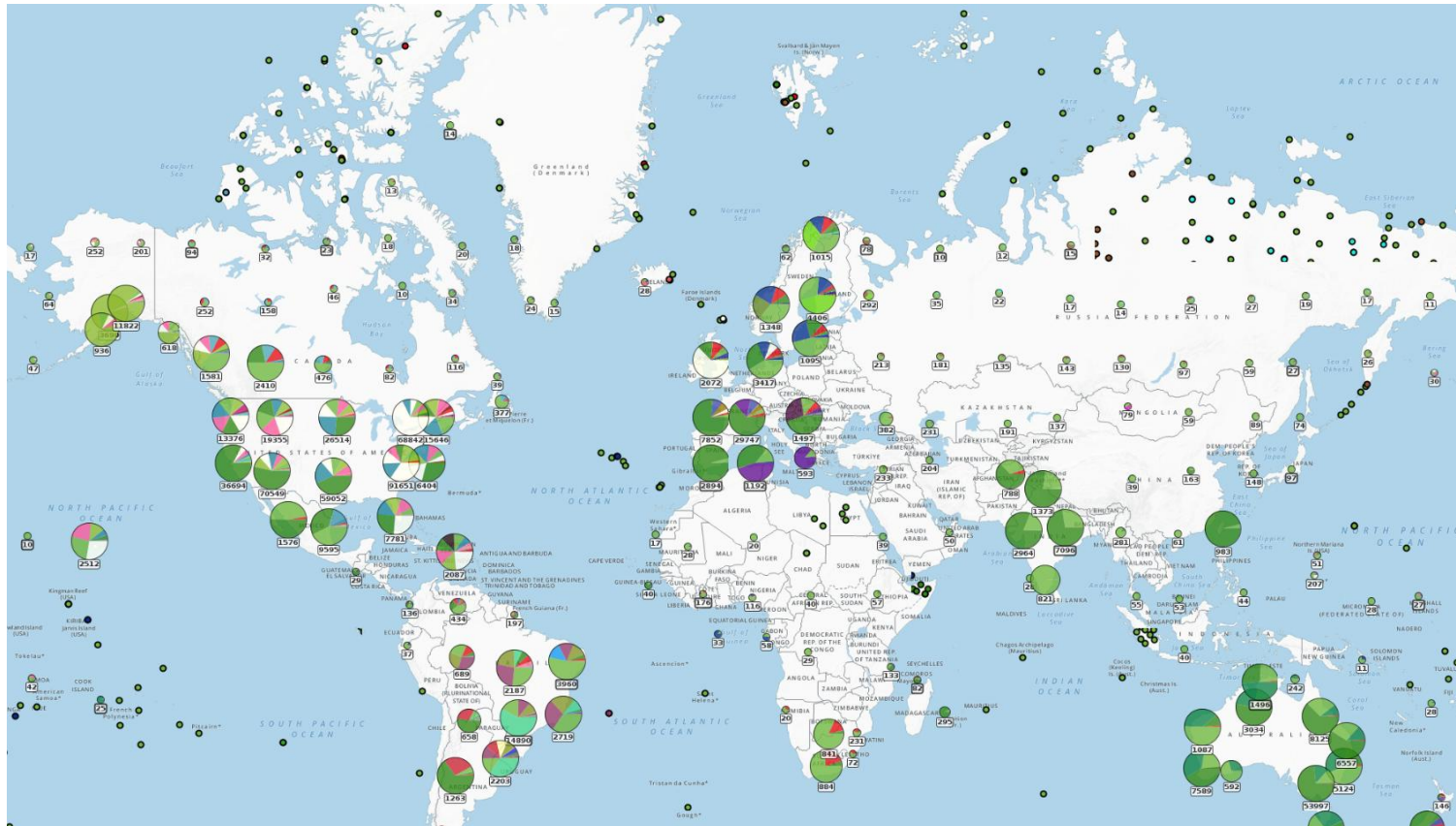
WHOS connects Hydrological Actors Worldwide

1. **Syntactic interoperability:** *Are the systems able to speak the same language, exchanging and understanding data with ease?*
2. **Semantic interoperability:** *Do the systems truly understand each other, sharing the same meaning behind every piece of data?*
3. **Spatial & temporal interoperability:** *Are location and time always in sync, no matter where or when the systems connect?*
4. **Technical/system interoperability:** *Can a diverse mix of databases, APIs, protocols, and applications seamlessly converse with one another?*



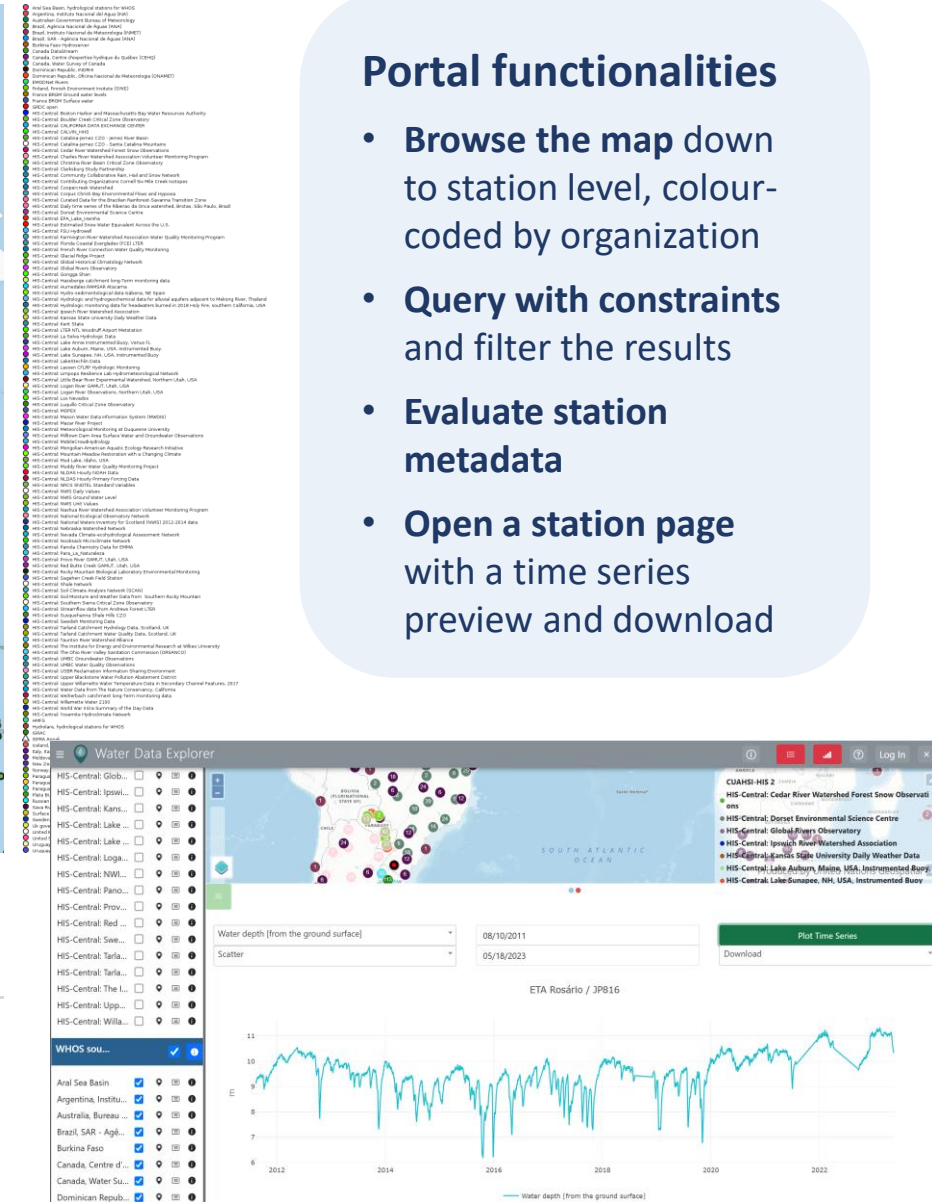
Multiple providers. Multiple users. Unified discovery and access.

WHOS Portal: Where is the Global Registry for Hydrology Data Providers



Portal functionalities

- Browse the map down to station level, colour-coded by organization
- Query with constraints and filter the results
- Evaluate station metadata
- Open a station page with a time series preview and download



<https://whos.geodab.eu/gs-service/whos/search.html>

130+



Participating institutions Time series

4.2M+

33+

Countries

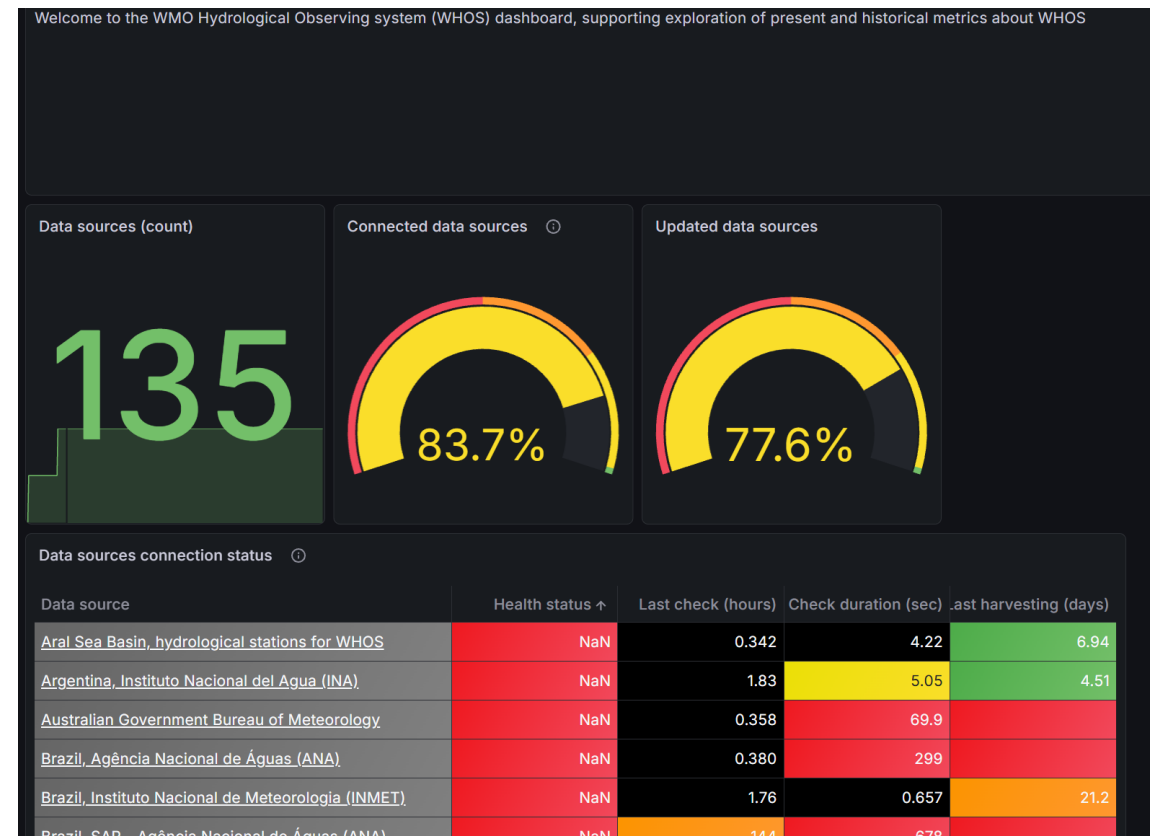
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Global data centres

WHOS Monitoring

- **Metadata quantity, quality and connection stability are monitored continuously**
- **Quantity** — stations and time series published by each provider
- **Quality** — completeness of the metadata records
- **Connection stability** — service availability and response over time
- **Reported overall and per data provider**

WHOS monitoring dashboard: connected and updated sources, plus connection status per provider



Two monitoring views: coverage by country, completeness by provider

Countries	Countries		
70	Country (ISO3) ↑	Source	Records
	AFG	GRDC open	973
	AFG	IGRAC	57
	ARG	GRDC open	907
	ARG	Argentina, Instituto Nacional del	5289
	AUS	GRDC open	18877
	AUS	IGRAC	67423
	AUS	Australian Government Bureau of	311616
	AUT	IGRAC	342
	AUT	GRDC open	3463
	AZE	Hydrolare, hydrological stations f	5
	BEL	IGRAC	1658
	BEL	GRDC open	1811

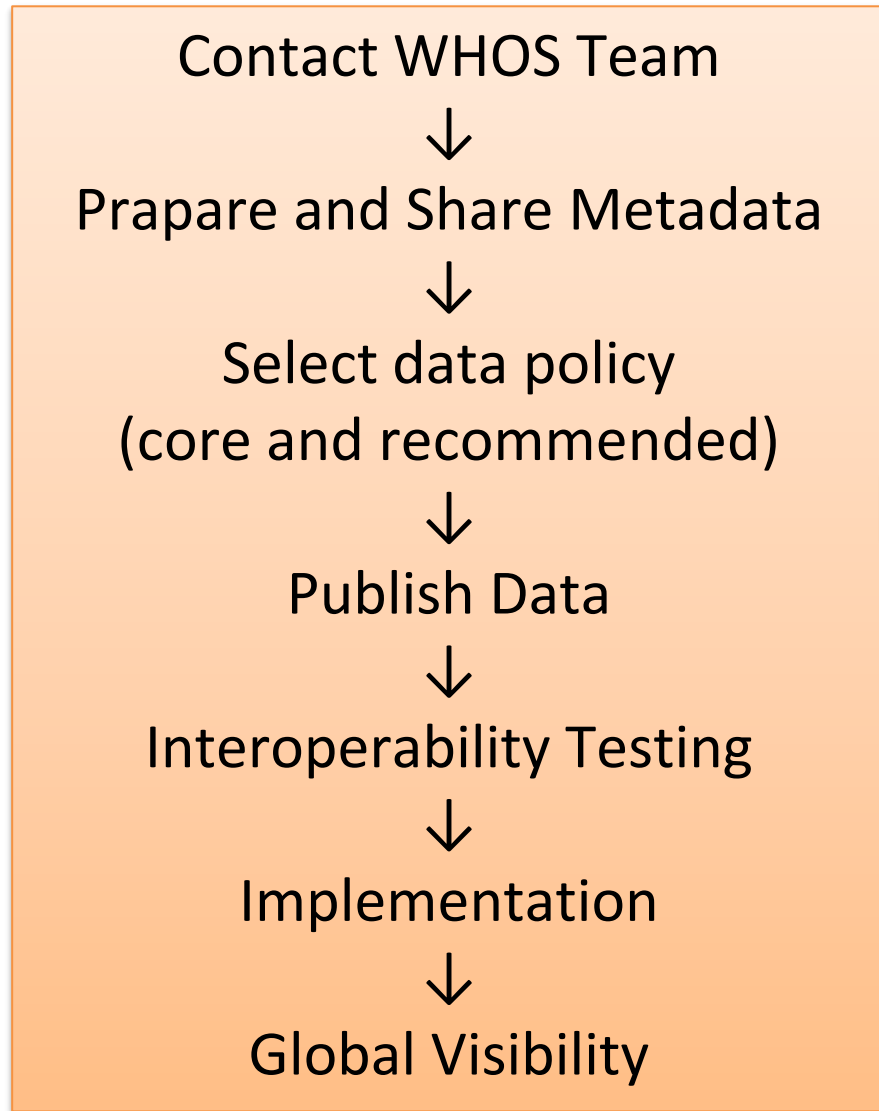
Monitoring dashboard: total countries and organizations by country

Metadata element completeness														
data source	title	abstract	bbox	time	country	parameter	units	parameter URI	platform	aggregation	interpolation	resolution	license	
Italy, Italian I	100	0	99.6	100	100	100	0	100	100	99.3	90.0	0	0	
Aral Sea Bas	100	100	0	100	0	100	100	100	100	100	100	100	0	
HMFS	100	100	100.0	100	0	100	0	0	100	0	0	0	0	
Argentina, In	100	0	100	100	100	100	100	100	100	41.7	91.2	41.7	0	
Australian G	100	100	88.7	100	100	100	0	14.1	100	78.7	100	78.7	0	
Burkina Faso	100	100	100	100	0	100	100	100	100	100	100	100	0	
Brazil, SAR -	100	100	100	0	100	100	64.4	100	100	13.8	100	0	0	
Brazil, Agênc	100	100	99.4	100	100	100	0	18.8	100	6.16	18.8	12.6	0	
Plata BUFR n	100	100	100	100	0	100	39.2	100	100	19.7	100	100	0	
Brazil, Institu	100	100	100	100	100	100	87.1	95.4	100	41.5	100	69.0	0	
Canada, Cen	100	0	95.9	100	100	100	0	100	99.8	0	0	0	0	
HIS-Central:	100	0	100	100	0	100	100	100	100	100	100	100	0	
HIS-Central:	100	0	100	100	0	100	100	87.7	100	64.2	84.0	64.2	0	
HIS-Central:	100	0	100	100	0	100	100	79.1	100	100	100	100	0	
HIS-Central:	100	0	100	100	0	100	100	72.7	100	6.06	15.2	6.06	0	

Metadata element completeness report, used to target improvements in metadata quality

Both views are refreshed continuously, so gaps can be traced to the provider that owns them.

Why and How: Joining WHOS



Benefits: Why Join WHOS



Global Visibility

Increase accessibility of hydrological datasets.



Collaboration

Enable data sharing across institutions and countries.



Standardization

Adopt common standards that reduce duplication.



Better Decisions

Support forecasting, planning, disaster risk reduction, and sustainable water management, better migration and conflict management, etc



Socio-Economic Impact

Hydrological data drives socio-economic impact by supporting better decisions



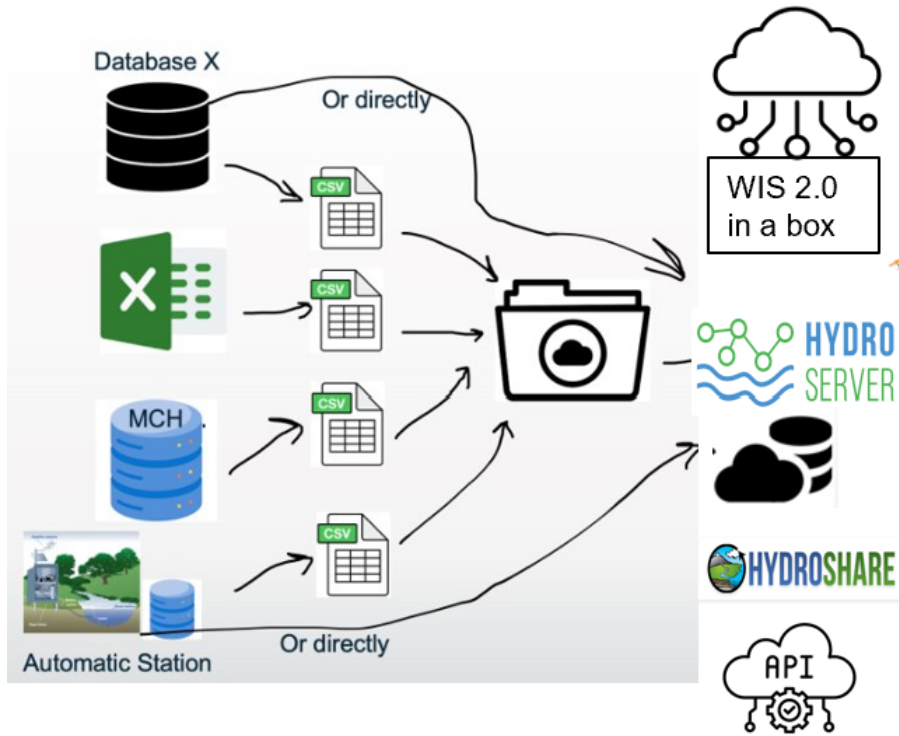
WMO OMM



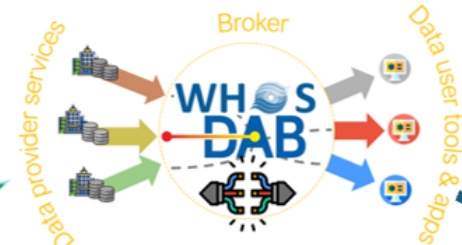
whos@wmo.int

Data Flow in WHOS ecosystem

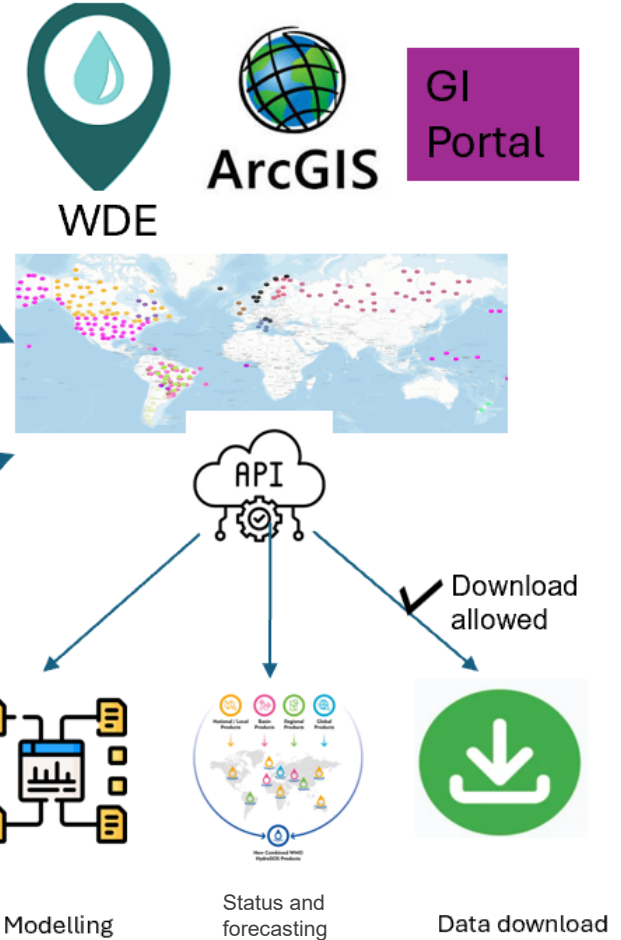
Data providers



Broker



Data consumers



Caches/ Archives



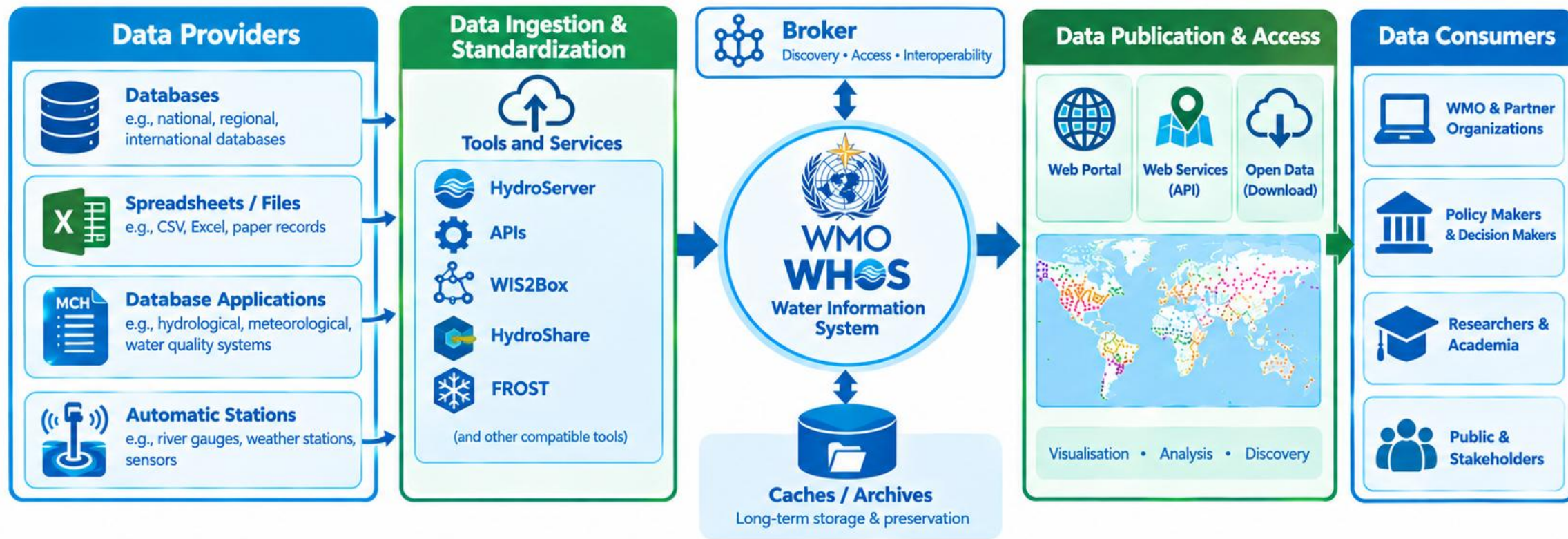
WHOS: Connecting Data Providers and Users. WHOS enables seamless discovery and access to hydrological data across multiple organizations.



WHOS does not act as a data warehouse. Data remain with the providers while becoming globally and regionally discoverable and accessible.

Data Flow in the WHOS Ecosystem

From global data providers to local and global users — for better water and climate decisions.



WHOS

Connecting data providers and users.
Enables seamless discovery and access
to hydrological data across multiple organizations.



Open & Interoperable

Uses international standards
(WIS 2.0) and open interfaces
(API, web services).



Reliable & Sustainable

Maintains global and regional
availability, with long-term storage
and preservation.



Better Decisions

Supports water security,
climate resilience and sustainable
development.

WHOS Implementation

Standardization

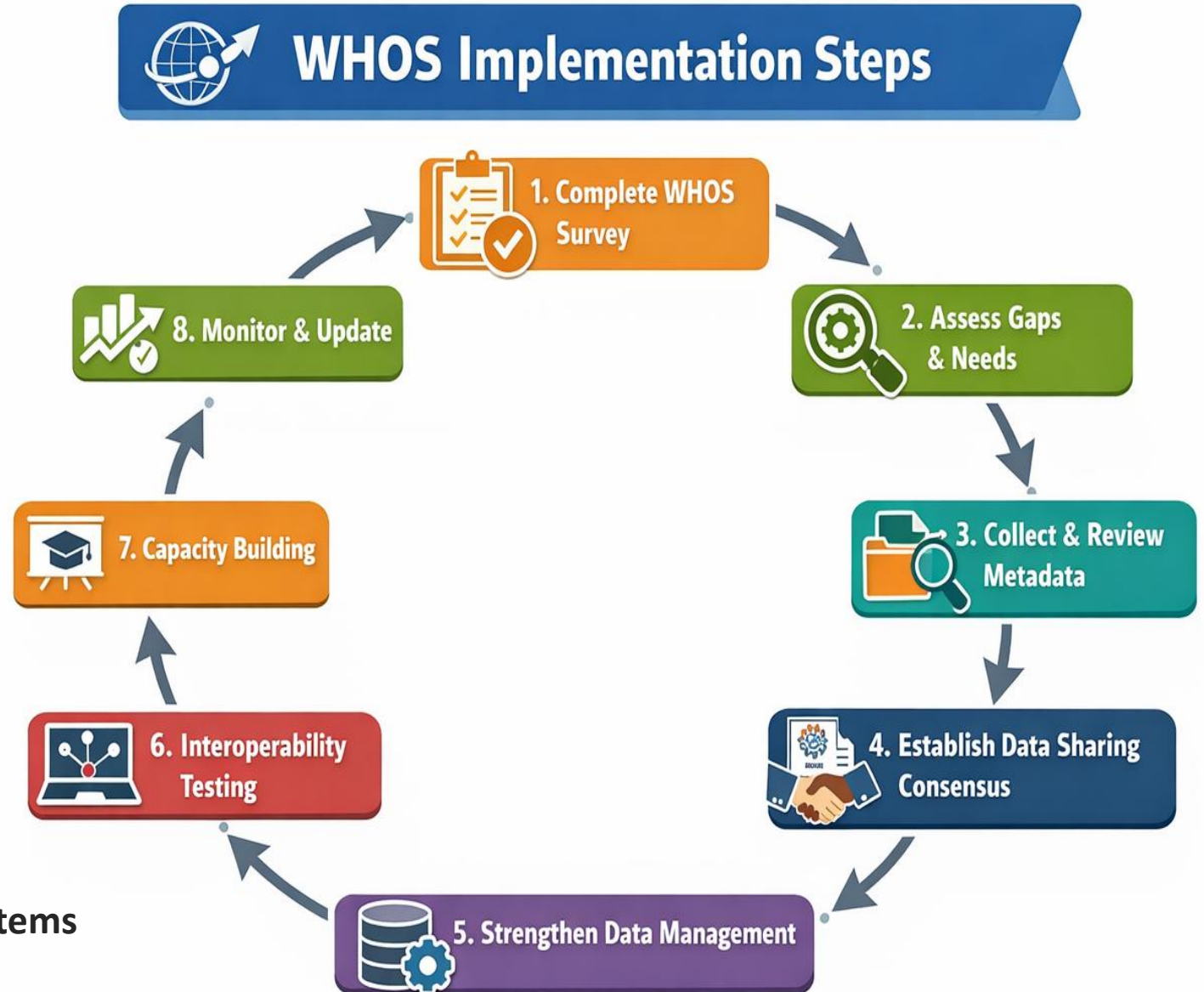
- Common metadata, data and protocol standards
- ISO 19115, WaterML, OGC W*S, THREDDS
- Cuts heterogeneity at the source, but does not remove it

+

Brokering approach

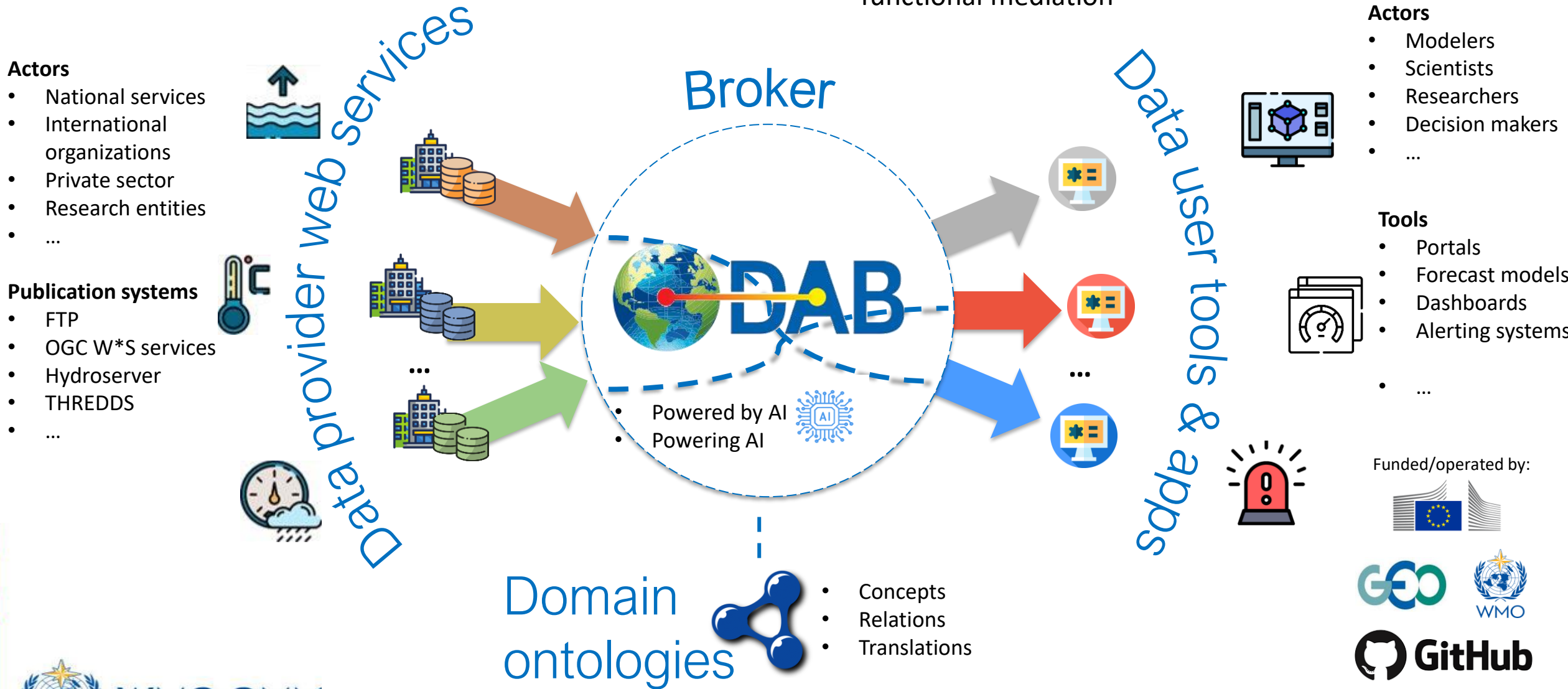
- A middle layer, the DAB, mediates what remains
- Data model mapping and functional mediation
- Providers and users keep the standards they already use

= Interoperability of hydrological systems



The brokering approach

Enables interoperability by introducing a middle layer component, the broker, dedicated to the harmonization through data model mapping and functional mediation



A selection of WHOS supported data publication systems



Hydroserver & HIS-Central (CUAHSI WOF)



Hydroserver2 (OGC STA)



FROST server (OGC STA)



MCH (MCH API)



52north SOS (OGC SOS)



Wis2box (OGC API record)



Custom APIs



SOAP/WSDL



Kisters



OData



and others ...
WMO OMM



Standardization

Tackles big data variety



Standardization provides guidance on implementation of:

Metadata: ISO 19115, Dublin Core, DIF, EML, ...

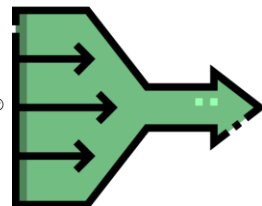
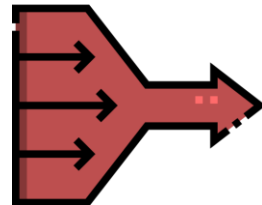
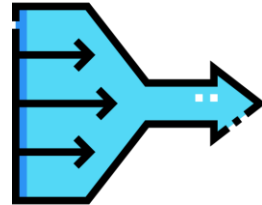
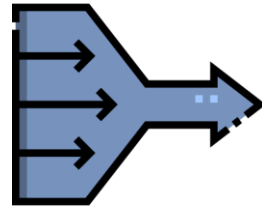
Data: NetCDF-CF, WaterML, GRIB, ...

Communication protocols: OGC W*S, THREDDS, WaterOneFlow, ...



Without standards:
High heterogeneity
of implementations

Interoperability is hard



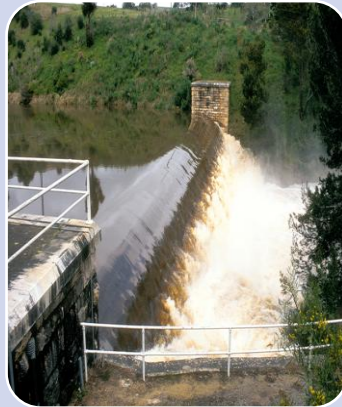
With standards:
Overall
heterogeneity is
reduced, but still
present

Interoperability is eased



WaterML2.0 suite of standards

- Implemented in WMO information system
- In many organizations: UNEP, USGS, US EPA, NrCan, BRGM, NIWA, AU BOM etc...
- And softwares : Kisters, 52°N, Aquatic Informatics, ...
- Regularly enhanced through partners feedback and Interoperability Experiments (the latest : 2022-2025 Water Quality IE n°1)



Part 1 – Water
Quantity
Timeseries

Being updated along
with TimeSeriesML 2.0
as per Water
Quality IE conclusions

Part 2 – Ratings,
Gaugings and
Sections

Part 3 – Surface
water features

Part 4 –
Groundwater

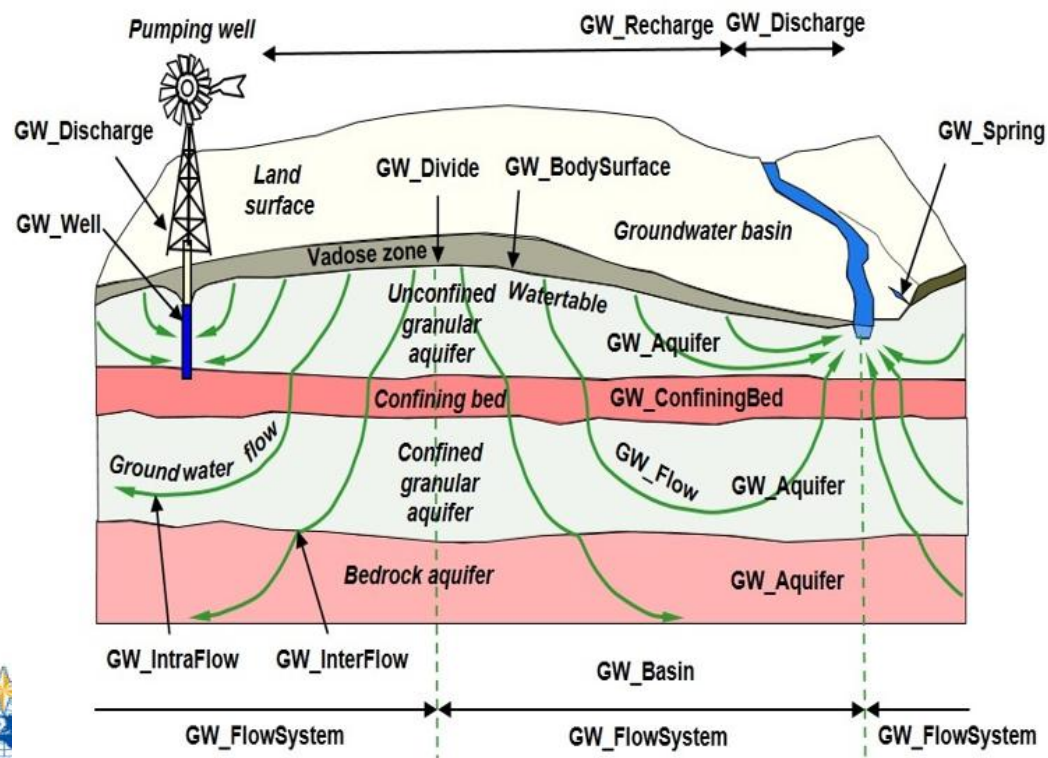
Part 5 –
Water quality
(best practice)

Will be updated after
TimeSeriesML 2.0



WaterML2.0: part 4 – GroundWater Markup Language 2 (GWML2)

Conceptual and logical model describing surface water hydrologic features such as rivers, lakes, catchments



- Hydrogeological units,
- Aquifers,
- Voids,
- Fluid bodies,
- Hydraulic conductivity,
- Water wells,
- Springs



WHOS and WMO Systems

WMO Metadata Registry i.e OSCAR



WIGOS — WHOS — WIS2



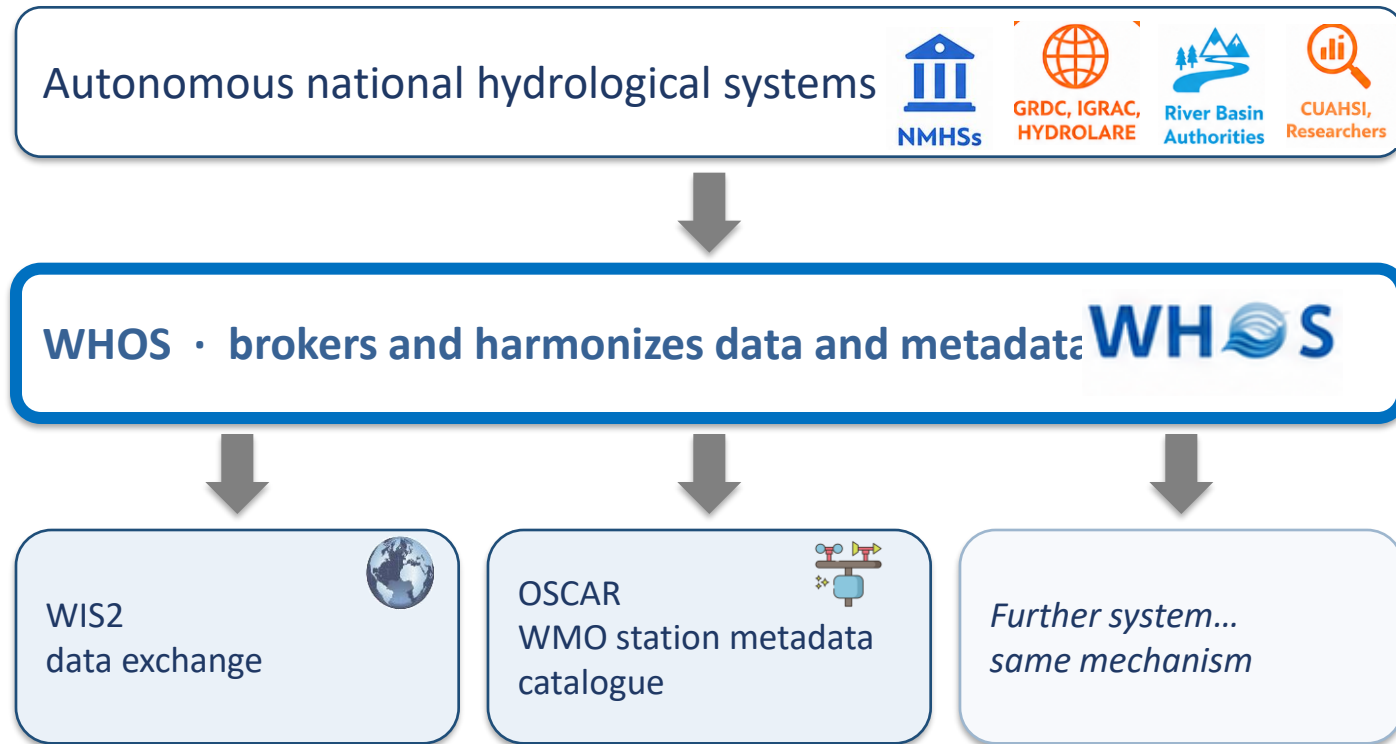
National Systems



Data Consumers

- **WHOS technologies complements existing WMO systems**
 - OSCAR, WIGOS, WIS2, and national systems.
- **Provide seamless access to hydrological information**

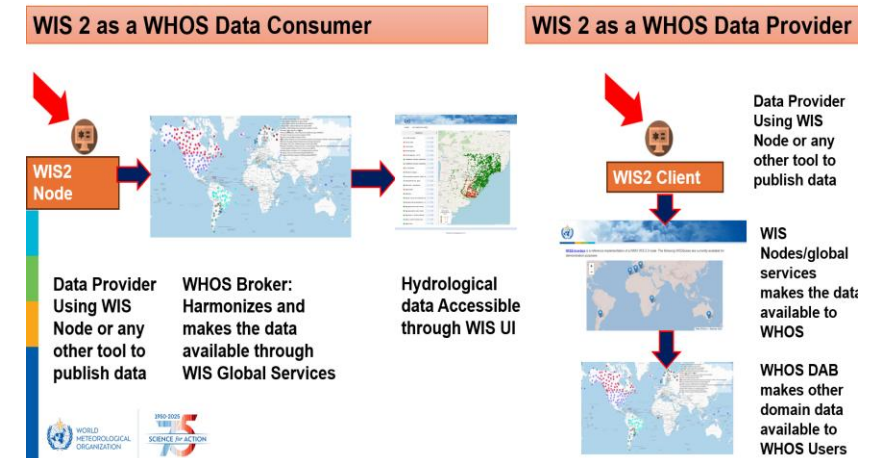
As with WIS, WHOS feeds national station metadata into OSCAR automatically



No extra work for the provider: stations published nationally appear in WIS, OSCAR or other relevant systems.



WHOS harvests station metadata and pushes it to OSCAR



The same mechanism already in place for WIS2

WHOS Further Development

The Future of Hydrological Data Sharing

Global Coverage

Increase participation of institutions.



Real-Time Exchange

Support operational decision-making.



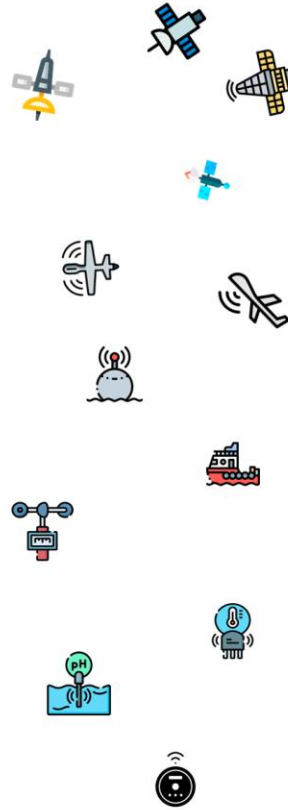
Strong Partnerships

Strengthen collaboration across sectors.



Better Decisions

Improve resilience through trusted information.



Volume

Quantity of generated and stored data

Variety

The type and nature of the data

Velocity

The speed at which the data is generated

Veracity

The truthfulness of reliability of the data

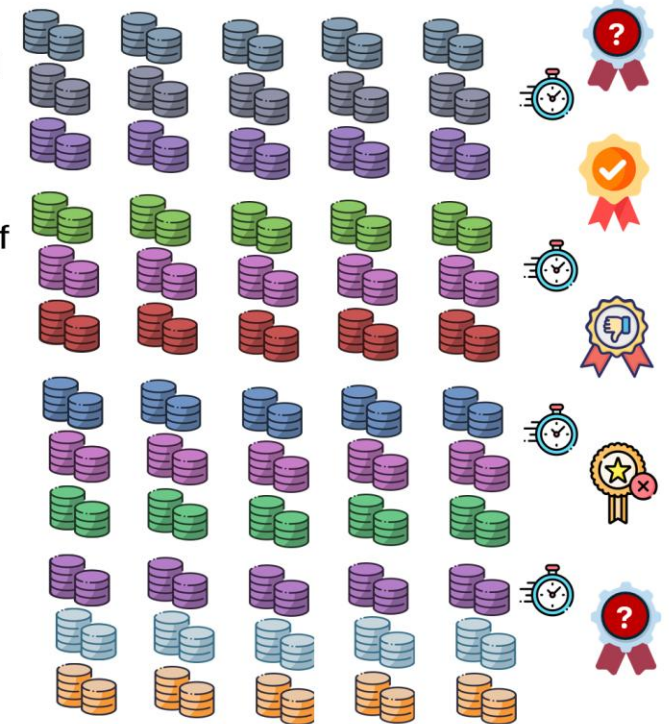


Table for Defining Core and Recommended Data –Initial List

Observations	Core	Recommended
Stream (Water) level	Daily and monthly	Shorter intervals
Lake/Reservoir level	Daily and monthly	Shorter intervals
Stream discharge/ Streamflow/Discharge	Daily	Shorter intervals
Lake/ Reservoir inflow	Daily and monthly	Shorter intervals
Lake/ Reservoir outflow	Daily and monthly	shorter intervals
Groundwater level	monthly, seasonal	Daily and shorter intervals
Snow depth	Daily, monthly	Shorter intervals
Water equivalent of snow cover	Weekly and monthly	Daily and shorter intervals
Soil moisture/soil moisture content	Monthly and seasonal	Hourly and daily
Evapotranspiration	Monthly and seasonal	Daily and shorter intervals
Precipitation (rain or snow) cumulative	Hourly and daily	Shorter intervals
Air temperature	Monthly, daily and hourly	Shorter intervals

Observations - Column A	Core	Recommended
Stream velocity/river velocity	x	Shorter intervals
Suspended Sediment discharge/transport	x	Daily and monthly
Water temperature	x	Shorter intervals, daily, and monthly
Groundwater Abstraction/ extraction	x	Daily, weekly, monthly, seasonal, and decade
Lake Ice Cover	x	Shorter intervals, Daily, weekly, and monthly
Lake Ice Thickness [cm]	x	Weekly and monthly
pH	x	Daily and monthly
Dissolved oxygen saturation	x	Weekly and monthly
Electrical conductivity (of water)	x	Shorter interval, daily, and monthly
Total dissolved solids	x	Weekly, monthly, and seasonal
Total dissolved Nitrogen	x	Weekly, monthly, and seasonal
Total dissolved Phosphorus	x	Weekly, monthly, and seasonal
Turbidity	x	Weekly, monthly, seasonal

Initial list of core and recommended data for hydrology wilbe submitted to INFCOM-4



WMO OMM

Thank you!

Let's work together to make hydrological data discoverable, accessible, and interoperable.

Experts, Contributors and Partners

For more details, contact: whos@wmo.int